Comprehensive Notes: Graphing Trigonometric Functions

Trigonometry Study Guide

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1 Standard Form of Trigonometric Functions

Any transformational sine or cosine function can be written in the general form:

$$y = A\sin(B(x - C)) + D \quad \text{or} \quad y = A\cos(B(x - C)) + D$$

Where each parameter dictates a specific geometric transformation on the parent graph:

- **Amplitude ($|A|$):** The vertical stretch or compression. Represents half the distance between the maximum and minimum values. If $A < 0$, the graph is reflected across the midline.
- **Period (T):** The distance required for the function to complete one full cycle.

$$\text{Period } (T) = \frac{2\pi}{|B|}$$

- **Phase Shift (C):** The horizontal displacement of the function.
  - If $C > 0$, the graph shifts to the **right** by C units.
  - If $C < 0$, the graph shifts to the **left** by $|C|$ units.
- **Vertical Shift / Midline (D):** The horizontal line $y = D$ around which the wave oscillates.

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2 Key Characteristics of Parent Functions

2.1 Sine and Cosine

2.2 Tangent and Reciprocal Functions

- **Tangent ($y = \tan(x)$):**

Property	**Sine Function** $y = \sin(x)$	**Cosine Function** $y = \cos(x)$
Domain	$(-\infty, \infty)$	$(-\infty, \infty)$
Range	$[-1, 1]$	$[-1, 1]$
Period	2π	2π
Amplitude	1	1
Symmetry	Odd (Symmetric about origin)	Even (Symmetric about y-axis)
Key Points (1 Cycle)	$(0,0), (\frac{\pi}{2}, 1), (\pi, 0), (\frac{3\pi}{2}, -1), (2\pi, 0)$	$(0,1), (\frac{\pi}{2}, 0), (\pi, -1), (\frac{3\pi}{2}, 0), (2\pi, 1)$

Table 1: Comparison of parent Sine and Cosine functions.

  - Period: $T = \frac{\pi}{|B|}$
  - Vertical Asymptotes: $x = \frac{\pi}{2} + k\pi$ for any integer k.
  - Domain: $x \neq \frac{\pi}{2} + k\pi$; Range: $(-\infty, \infty)$.

- **Cosecant ($y = \csc(x) = \frac{1}{\sin(x)}$):**
  - Period: 2π; Vertical Asymptotes at $x = k\pi$.
  - Local minima/maxima touch the peak and trough points of $y = \sin(x)$.
- **Secant ($y = \sec(x) = \frac{1}{\cos(x)}$):**
  - Period: 2π; Vertical Asymptotes at $x = \frac{\pi}{2} + k\pi$.

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3 Step-by-Step Graphing Strategy

To accurately sketch a function such as $y = A\sin(B(x - C)) + D$:

1. **Identify Core Parameters:** Extract A, B, C, and D from the given equation (factor out B inside the argument if necessary).
2. **Determine Key Attributes:**
   - Midline: $y = D$
   - Maximum Value: $y_{\text{max}} = D + |A|$
   - Minimum Value: $y_{\text{min}} = D - |A|$
   - Period: $T = \frac{2\pi}{|B|}$
   - Increment (Quarter-Period): $\Delta x = \frac{T}{4}$
3. **Calculate Key X-Values:** Start at the phase shift $x_0 = C$, and consecutively add the quarter-period increment Δx:

$$x_0 = C, \quad x_1 = C + \Delta x, \quad x_2 = C + 2\Delta x, \quad x_3 = C + 3\Delta x, \quad x_4 = C + 4\Delta x$$

4. **Evaluate Corresponding Y-Values:**

- **Sine Pattern:** Midline $\rightarrow$ Max $\rightarrow$ Midline $\rightarrow$ Min $\rightarrow$ Midline
- **Cosine Pattern:** Max $\rightarrow$ Midline $\rightarrow$ Min $\rightarrow$ Midline $\rightarrow$ Max
- *(Note: If $A < 0$, invert the Max and Min points.)*

5. **Plot and Smooth Curve:** Plot the five primary points, draw vertical asymptotes if applicable, and connect points with a continuous smooth wave.

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4 Worked Example

Problem: Graph the function $y = -3\cos\left(2x - \frac{\pi}{2}\right) + 1$ over one full period.

Solution Walkthrough

Step 1: Rewrite in factored standard form

$$y = -3\cos\left(2\left(x - \frac{\pi}{4}\right)\right) + 1$$

Step 2: Identify parameters

- $A = -3$ (Amplitude = 3, reflected over midline)
- $B = 2 \implies$ Period $T = \frac{2\pi}{2} = \pi$
- $C = \frac{\pi}{4}$ (Phase shift right by $\frac{\pi}{4}$)
- $D = 1$ (Midline at $y = 1$)

Step 3: Calculate increments and key points

- Quarter-Period Increment: $\Delta x = \frac{T}{4} = \frac{\pi}{4}$
- $x_0 = \frac{\pi}{4}$
- $x_1 = \frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2}$
- $x_2 = \frac{\pi}{2} + \frac{\pi}{4} = \frac{3\pi}{4}$
- $x_3 = \frac{3\pi}{4} + \frac{\pi}{4} = \pi$
- $x_4 = \pi + \frac{\pi}{4} = \frac{5\pi}{4}$

Step 4: Determine point coordinates

Since $A = -3 < 0$, the inverted cosine pattern is: **Min $\rightarrow$ Midline $\rightarrow$ Max $\rightarrow$ Midline $\rightarrow$ Min**.

x	y	**Key Feature**
$\frac{\pi}{4}$	$1-3=-2$	Minimum point
$\frac{\pi}{2}$	1	Midline intercept
$\frac{3\pi}{4}$	$1+3=4$	Maximum point
π	1	Midline intercept
$\frac{5\pi}{4}$	$1-3=-2$	Minimum point

Table 2: Key coordinates for $y=-3\cos\left(2\left(x-\frac{\pi}{4}\right)\right)+1$ over one period.